Homework 2

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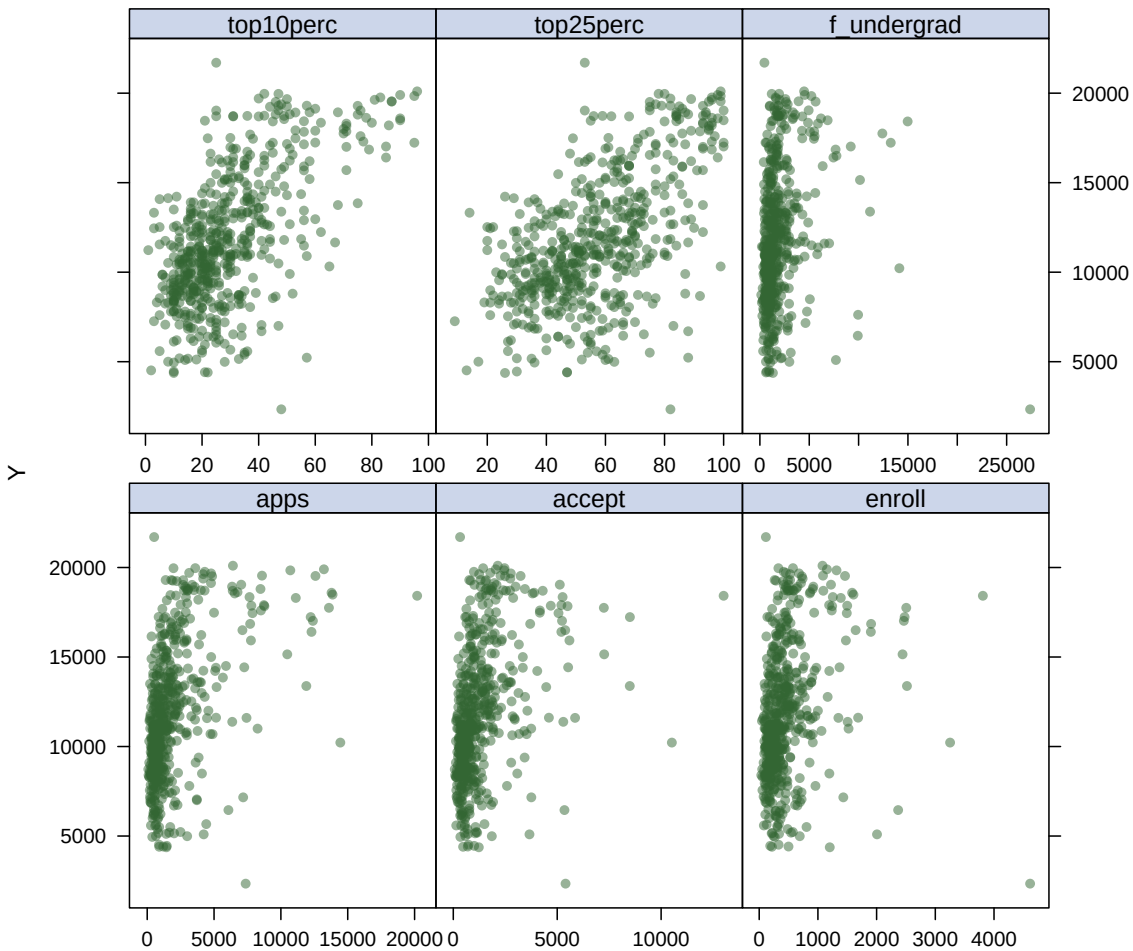
Initialize Data

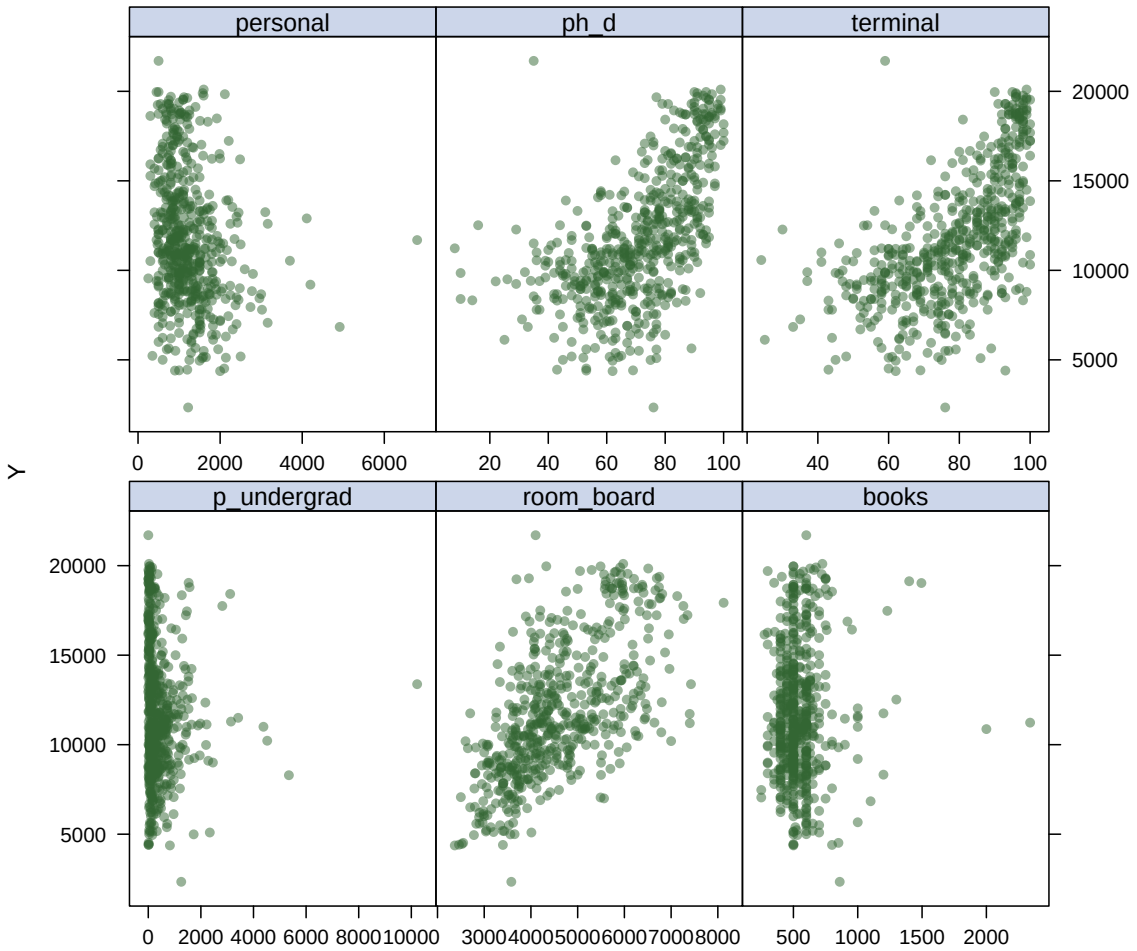
```r
college <- read.csv("data/College.csv",  na = c("NA", ".", "")) |>
  janitor::clean_names() |>
  select(-college) |>
  select(outstate, everything()) |>
  mutate(across(everything(), as.numeric))

summary(college) # check overall structure
```

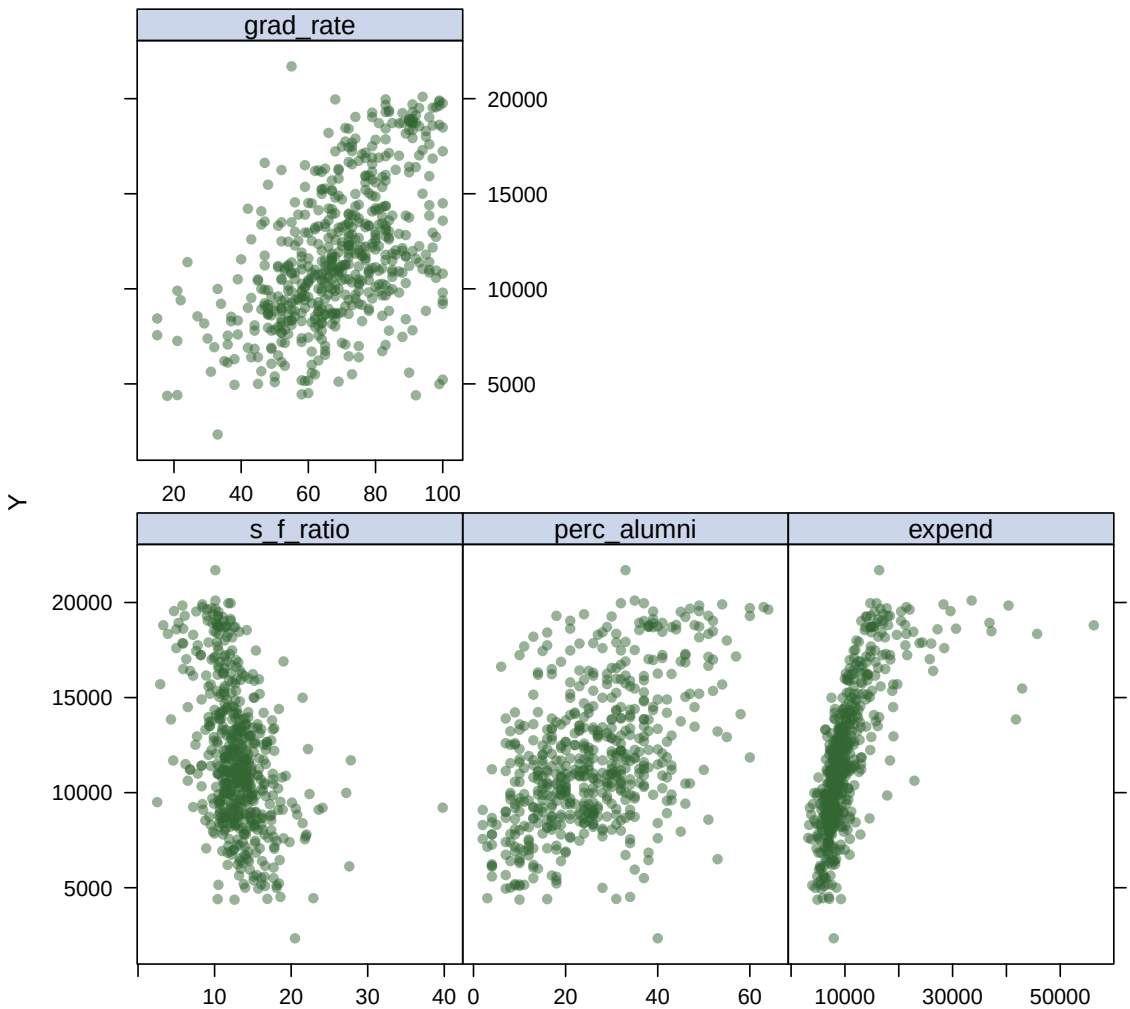
```
##     outstate          apps           accept          enroll
##  Min.   : 2340   Min.   :   81   Min.   :   72   Min.   :  35.0
##  1st Qu.: 9100   1st Qu.:  619   1st Qu.:  501   1st Qu.: 206.0
##  Median :11200   Median : 1133   Median :  859   Median : 328.0
##  Mean   :11802   Mean   : 1978   Mean   : 1306   Mean   : 456.9
##  3rd Qu.:13970   3rd Qu.: 2186   3rd Qu.: 1580   3rd Qu.: 520.0
##  Max.   :21700   Max.   :20192   Max.   :13007   Max.   :4615.0
##    top10perc       top25perc       f_undergrad     p_undergrad
##  Min.   : 1.00   Min.   :  9.00   Min.   :  139   Min.   :    1
##  1st Qu.:17.00   1st Qu.: 42.00   1st Qu.:  840   1st Qu.:   63
##  Median :25.00   Median : 55.00   Median : 1274   Median :  207
##  Mean   :29.33   Mean   : 56.96   Mean   : 1872   Mean   :  434
##  3rd Qu.:36.00   3rd Qu.: 70.00   3rd Qu.: 2018   3rd Qu.:  541
##  Max.   :96.00   Max.   :100.00   Max.   :27378   Max.   :10221
##    room_board       books           personal         ph_d
##  Min.   :2370   Min.   : 250.0   Min.   : 250   Min.   :  8.00
##  1st Qu.:3736   1st Qu.: 450.0   1st Qu.: 800   1st Qu.: 60.00
##  Median :4400   Median : 500.0   Median :1100   Median : 73.00
##  Mean   :4586   Mean   : 547.5   Mean   :1214   Mean   : 71.09
##  3rd Qu.:5400   3rd Qu.: 600.0   3rd Qu.:1500   3rd Qu.: 85.00
##  Max.   :8124   Max.   :2340.0   Max.   :6800   Max.   :100.00
##     terminal        s_f_ratio      perc_alumni        expend        grad_rate
##  Min.   : 24.00   Min.   : 2.50   Min.   : 2.00   Min.   : 3186   Min.   : 15
##  1st Qu.: 68.00   1st Qu.:11.10   1st Qu.:16.00   1st Qu.: 7477   1st Qu.: 58
##  Median : 81.00   Median :12.70   Median :25.00   Median : 8954   Median : 69
##  Mean   : 78.53   Mean   :12.95   Mean   :25.89   Mean   :10486   Mean   : 69
##  3rd Qu.: 92.00   3rd Qu.:14.50   3rd Qu.:34.00   3rd Qu.:11625   3rd Qu.: 81
##  Max.   :100.00   Max.   :39.80   Max.   :64.00   Max.   :56233   Max.   :118
```

```r
# we should cap graduation rate at 100%
college$grad_rate[college$grad_rate > 100] <- 100
college <- na.omit(college)

# matrix of predictors
```

```r
x <- model.matrix(outstate ~ ., college)[, -1]
# vector of response
y <- college$outstate

# Get the image parameters of the current trellis graph
theme1 <- trellis.par.get()
theme1$plot.symbol$col <- rgb(.2, .4, .2, .5)
theme1$plot.symbol$pch <- 16
theme1$plot.line$col <- rgb(.8, .1, .1, 1)
theme1$plot.line$lwd <- 2
theme1$strip.background$col <- rgb(.0, .2, .6, .2)
# Set the modified parameters as the global style for the trellis graph
trellis.par.set(theme1)
featurePlot(x, y, plot = "scatter", labels = c("", "Y"),
type = c("p"), layout = c(3, 2))
```




personal
ph_d
terminal
p_undergrad
room_board
books
Y
20000
15000
10000
5000
0
2000
4000
6000
8000
10000
20
40
60
80
100
3000
4000
5000
6000
7000
8000
500
1000
1500
2000



Partition the dataset into two parts: training data (80%) and test data (20%).

```
set.seed(123) # make this reproducible
data_split <- initial_split(college, prop = 0.8)
train_data <- training(data_split)
test_data  <- testing(data_split)
```

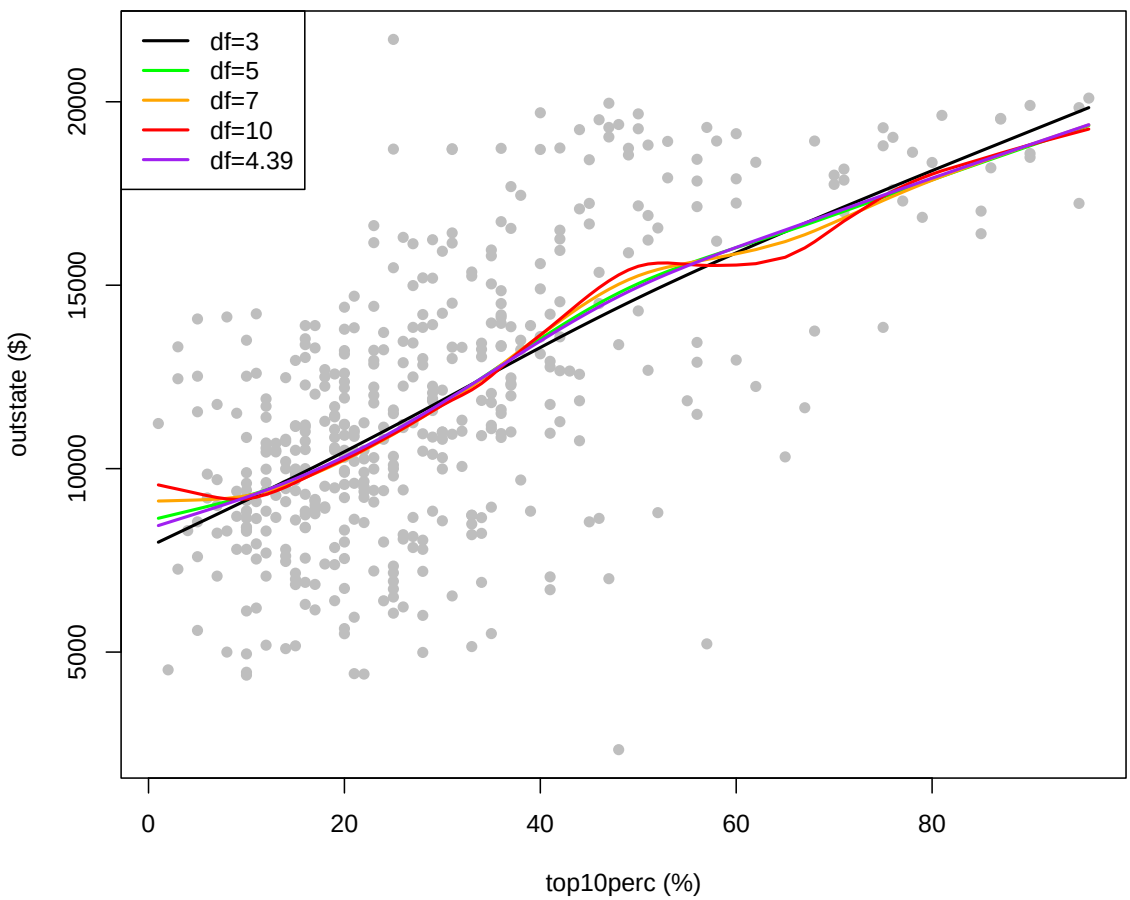
(a) Fit smoothing spline models to predict out-of-state tuition (Outstate) using Top10perc as the only predictor, across a range of degrees of freedom. Plot the fitted curve for each degree of freedom. Describe the patterns you observe as the degrees of freedom change. Choose an appropriate degree of freedom for the model and plot this optimal fit. State the criteria you used to select the degree of freedom.

```
set.seed(123)
pred.grid <- seq(from = -20, to = 120, by = 1)

# predictor and response matrices
x <- train_data$top10perc
y <- train_data$outstate

# smoothing splines with different df
fit3  <- smooth.spline(x, y, df = 3)
fit5  <- smooth.spline(x, y, df = 5)
fit7  <- smooth.spline(x, y, df = 7)
```

```r
fit10 <- smooth.spline(x, y, df = 10)
fit_opt <- smooth.spline(x, y)
fit_opt$df # optimal df is 4.394647
```

```
## [1] 4.394647
```

```r
plot(x, y,
     col = "grey",
     pch = 16,
     xlab = "top10perc (%)",
     ylab = "outstate ($)")

lines(predict(fit3), col = "black", lwd = 2)
lines(predict(fit5), col = "green", lwd = 2)
lines(predict(fit7), col = "orange", lwd = 2)
lines(predict(fit10), col = "red", lwd = 2)
lines(predict(fit_opt), col = "purple", lwd = 2)

legend("topleft",
       legend = c("df=3","df=5","df=7","df=10", "df=4.39"),
       col = c("black","green","orange","red", "purple"),
       lwd = 2)
```

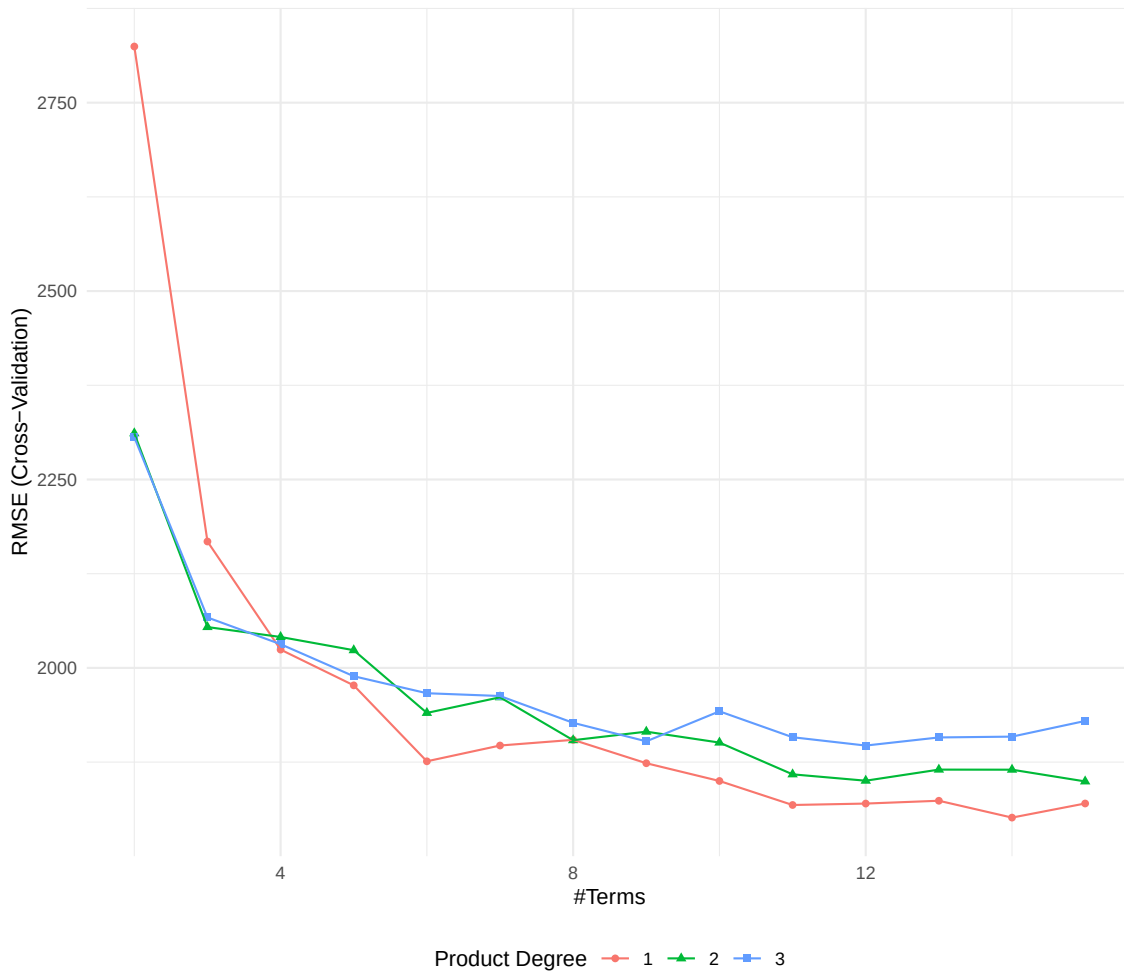


As the df increases from 3 to 10, the natural spline gets more flexible, resulting in overfitting. As determined by the optimal smoothing spline, the appropriate degree of freedom was 4.39 (generalized cross-validation is used to select the degree of freedom or the trace of the smoother matrix).

(b) Train a multivariate adaptive regression spline (MARS) model to predict the response variable. Report the regression function. Present the partial dependence plot of an arbitrary predictor in your model. Report the test error.

```r
set.seed(123)

# matrix of predictors
x <- model.matrix(outstate ~ ., train_data)[, -1]
# vector of response
y <- train_data$outstate

ctrl1 <- trainControl(method = "cv", number = 10) # 10-fold cv

mars_grid <- expand.grid(degree = 1:3, nprune = 2:15)

mars.fit <- train(x, y,
  method = "earth",
  tuneGrid = mars_grid,
  trControl = ctrl1)
  ggplot(mars.fit)
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the caret package.
##   Please report the issue at <https://github.com/topepo/caret/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Let's try to get a sense of the selected hyperparameters

```
mars.fit$bestTune # nprune = 14, df = 1
```

```
##    nprune degree
## 13     14      1
```

```
coef(mars.fit$finalModel)
```

```
##          (Intercept)     h(expend-15954)     h(85-grad_rate)  h(4440-room_board)
##        12974.7011817          -0.7523615         -25.1664857          -1.3090278
## h(f_undergrad-1365) h(1365-f_undergrad)   h(22-perc_alumni)        h(apps-1377)
##           -1.0904120          -2.2469701         -83.5697756           0.3613673
##        h(enroll-213)       h(213-enroll)      h(expend-6951) h(f_undergrad-2993)
##           -2.9673230          13.5624272           0.7449182           1.1719119
##       h(accept-1540)      h(1540-accept)
##            0.4147319          -1.8564357
```

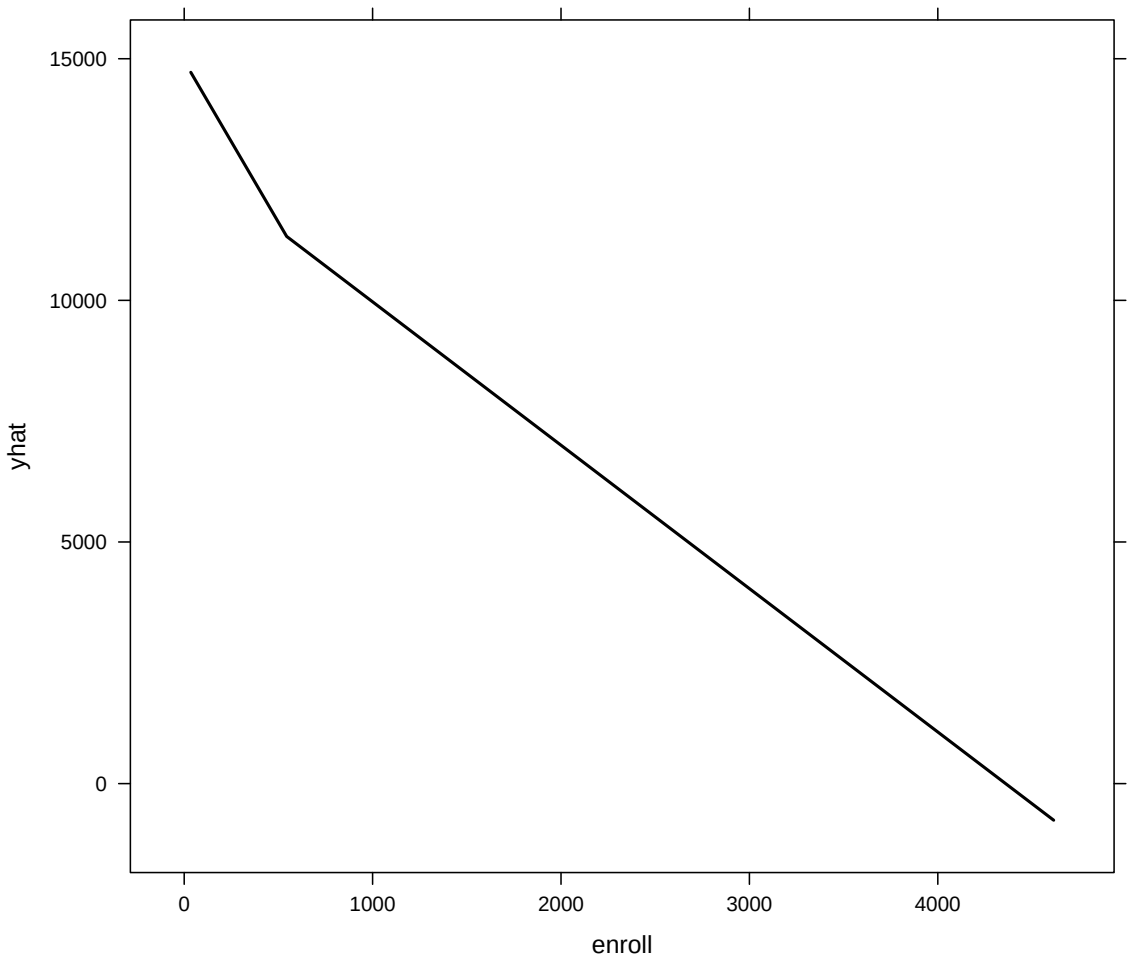
```
print(mars.fit$finalModel)
```

```
## Selected 14 of 30 terms, and 8 of 16 predictors (nprune=14)
## Termination condition: Reached nk 33
## Importance: expend, room_board, perc_alumni, accept, f_undergrad, apps, ...
## Number of terms at each degree of interaction: 1 13 (additive model)
## GCV 3017103    RSS 1205673245    GRSq 0.7914247    RSq 0.8147801
```

The regression function is

$$
\begin{aligned}
\hat{y} = 12974.7011817 \\
&- 0.7523615 \cdot h(\text{expend} - 15954) \\
&- 25.1664857 \cdot h(85 - \text{grad_rate}) \\
&- 1.3090278 \cdot h(4440 - \text{room_board}) \\
&- 1.0904120 \cdot h(\text{f_undergrad} - 1365) \\
&- 2.2469701 \cdot h(1365 - \text{f_undergrad}) \\
&- 83.5697756 \cdot h(22 - \text{perc_alumni}) \\
&+ 0.3613673 \cdot h(\text{apps} - 1377) \\
&- 2.9673230 \cdot h(\text{enroll} - 213) \\
&+ 13.5624272 \cdot h(213 - \text{enroll}) \\
&+ 0.7449182 \cdot h(\text{expend} - 6951) \\
&+ 1.1719119 \cdot h(\text{f_undergrad} - 2993) \\
&+ 0.4147319 \cdot h(\text{accept} - 1540) \\
&- 1.8564357 \cdot h(1540 - \text{accept})
\end{aligned}
$$

```r
p1 <- pdp::partial(mars.fit, pred.var = "enroll", grid.resolution = 10) |>
  pdp::plotPartial()

p1
```



This is the partial dependence plot on the estimated response variable with the predictor `enroll`. Increasing enrollment decreases the predicted out of state tuition.

```
mars_preds <- as.numeric(predict(mars.fit, newdata = test_data))
test_mse <- mean((test_data$outstate - mars_preds)^2, na.rm = TRUE)
print(paste("MARS Test MSE:", round(test_mse, 2)))
```

```
## [1] "MARS Test MSE: 2589498.05"
```

```
print(mars.fit$finalModel)
```

```
## Selected 14 of 30 terms, and 8 of 16 predictors (nprune=14)
## Termination condition: Reached nk 33
## Importance: expend, room_board, perc_alumni, accept, f_undergrad, apps, ...
## Number of terms at each degree of interaction: 1 13 (additive model)
## GCV 3017103    RSS 1205673245    GRSq 0.7914247    RSq 0.8147801
```

The MARS test MSE on the response variable `outstate` was 2,589,498.05. MARS selected 14 of 30 terms, and 8 of 16 predictors (nprune=14).

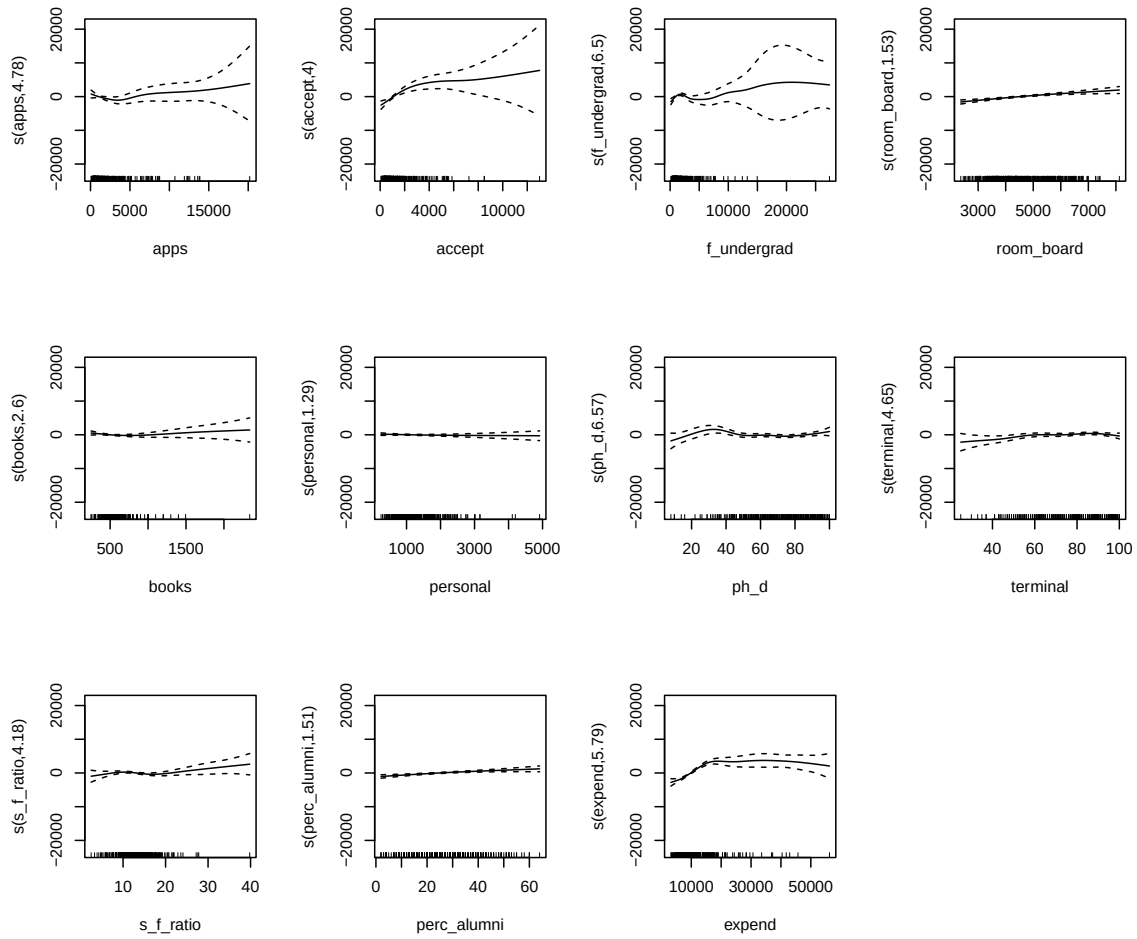
(c) Construct a generalized additive model (GAM) to predict the response variable. For the nonlinear terms included in your model, generate plots to visualize these relationships and discuss your observations. Report the test error.

```
set.seed(123)

gam.mod <- gam(
  outstate ~ s(apps) + s(accept) + s(enroll) + s(top10perc) + s(top25perc) +
    s(f_undergrad) + s(p_undergrad) + s(room_board) + s(books) +
    s(personal) + s(ph_d) + s(terminal) + s(s_f_ratio) +
    s(perc_alumni) + s(expend) + s(grad_rate),
  data = train_data
)

# Model summary
gam_sum <- summary(gam.mod)

# Find smooth terms (edf > 1)
nonlinear_idx <- which(gam_sum$s.table[, "edf"] > 1.001)

# See the smooth terms
rownames(gam_sum$s.table)[nonlinear_idx]
```

```
##  [1] "s(apps)"        "s(accept)"      "s(f_undergrad)" "s(room_board)"
##  [5] "s(books)"       "s(personal)"    "s(ph_d)"        "s(terminal)"
##  [9] "s(s_f_ratio)"   "s(perc_alumni)" "s(expend)"
```

The smooth or nonlinear terms are `apps`, `accept`, `f_undergrad`, `room_board`, `books`, `personal`, `ph_d`, `terminal`, `s_f_ratio`, `perc_alumni`, and `expend`.

```
# Plot only nonlinear / smooth terms
par(mfrow = c(3, 4))
for (i in nonlinear_idx) {
  plot(gam.mod,
       select = i,
       shade = TRUE,
       seWithMean = TRUE)
}
```



The terms `apps`, `accept`, `f_undergrad`, `room_board`, `ph_d`, `s_f_ratio`, `perc_alumni`, and `expend` have the most nonlinear relationships with the response variable `outstate`.

```
set.seed(123)

# MSE
gam.pred = predict(gam.mod, newdata = test_data)
y2 = test_data$outstate
mean((y2 - gam.pred)^2)
```

```
## [1] 2298158
```

The test MSE for the GAM model is 2,298,158.

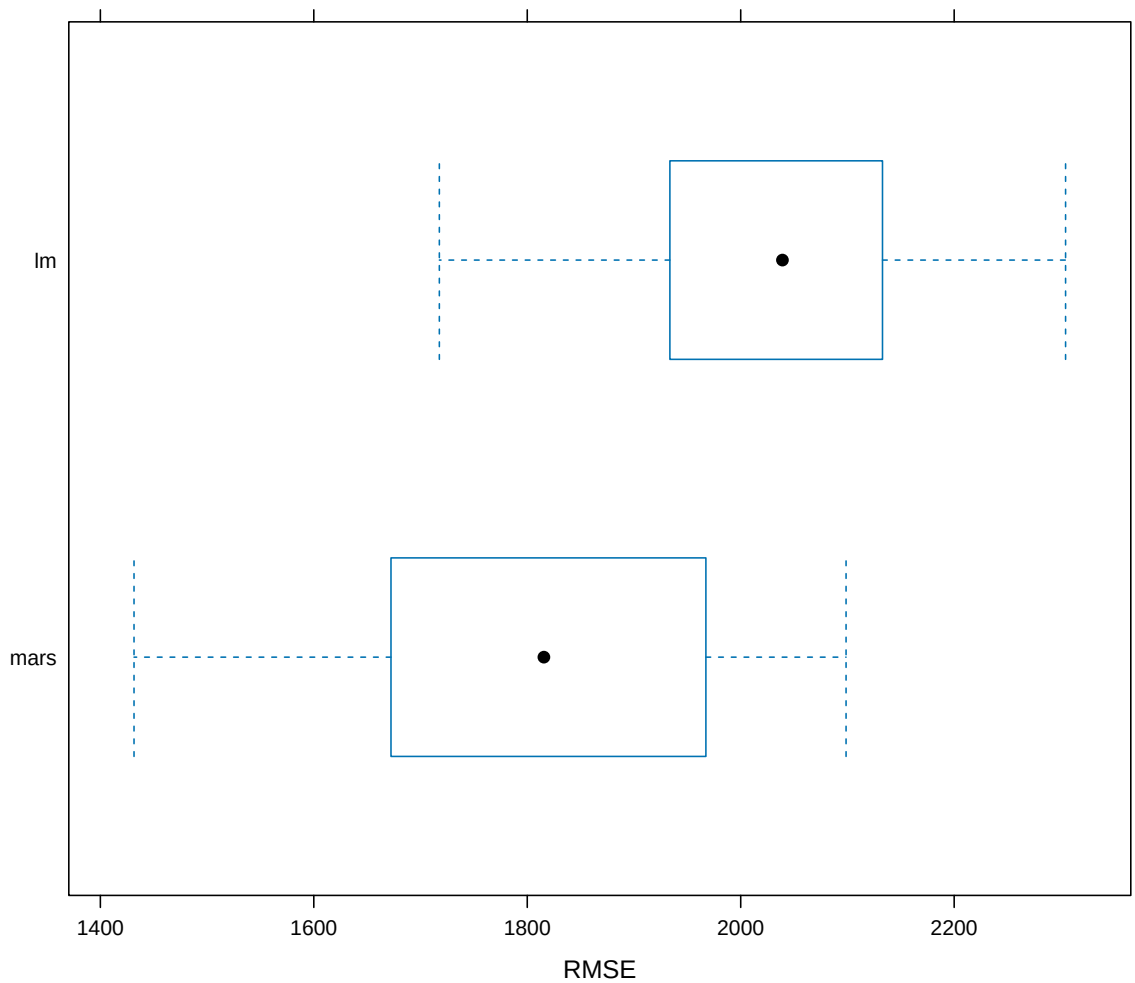
(d) In this dataset, would you favor a MARS model over a linear model for predicting out-ofstate tuition? Provide a clear justification for your choice.

```
set.seed(123)

lm.fit <- train(
  outstate ~ .,
  train_data,
  method = "lm",
  trControl = ctrl1
)
```

```r
# compare cross-validated results
resamp <- resamples(list(
  mars = mars.fit,
  lm   = lm.fit
))

summary(resamp)
```

```
##
## Call:
## summary.resamples(object = resamp)
##
## Models: mars, lm
## Number of resamples: 10
##
## MAE
##          Min.  1st Qu.   Median     Mean  3rd Qu.     Max. NA's
## mars 1122.963 1331.528 1415.798 1379.948 1479.699 1544.807    0
## lm   1461.167 1521.205 1581.671 1594.883 1690.479 1746.209    0
##
## RMSE
##          Min.  1st Qu.   Median     Mean  3rd Qu.     Max. NA's
## mars 1431.539 1697.234 1815.586 1801.349 1940.949 2098.725    0
## lm   1717.625 1958.750 2039.110 2024.479 2115.530 2304.403    0
##
## Rsquared
##           Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## mars 0.7066849 0.7727172 0.8037412 0.7867751 0.8082816 0.8275417    0
## lm   0.6969411 0.7218820 0.7369726 0.7335531 0.7456801 0.7679974    0
```

```r
# boxplot of RMSE
bwplot(resamp, metric = "RMSE")
```




```r
# evaluate the test MSE for linear model
lm.pred <- predict(lm.fit, newdata = test_data)
lm_mse <- mean((test_data$outstate - lm.pred)^2)
print(paste("Linear Model Test MSE:", round(lm_mse, 2)))
```

```
## [1] "Linear Model Test MSE: 3318588.69"
```

In this data I would favor the multivariate adaptive regression spline (MARS) model. The linear model has a test MSE of 3,318,588.69 compared to the MARS's test MSE of 2,589,498.05. The fact that the linear model performs worse on the testing (and training data makes sense); the MARS model can account for more nonlinearity in the relationship between the response variable `outstate` and the predictors.

(e) In practical applications, would you consider a MARS model to be superior to a linear model, or the other way around? Please justify your answer with appropriate reasoning.

If it is unjustified to believe that there is nonlinearity in the relationship between the response variable and the predictors, or if the linear model provides similar test error, a linear model is probably the better choice on the basis of interpretability. The multivariate adaptive regression spline (MARS) model is more helpful when the the data show curvature or threshold effects, because MARS models build a piecewise linear model using hinge basis functions and will automatically capture nonlinear predictor-response relationships.